Series Inductance Design in Isolated DC-DC Converters: DAB vs. CLLC

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1 Introduction

Both the Dual Active Bridge (DAB) and the CLLC resonant converter are bidirectional, galvanically isolated DC-DC topologies widely used in energy-storage interfaces, on-board chargers, and solid-state transformer applications. In both cases a series inductance L_s—realised either as the leakage inductance of the transformer or as an external component—plays a central role in limiting and controlling the power flow. The underlying physics, however, differs substantially between the two topologies, leading to different design equations and different degrees of freedom for the power-stage engineer.

This note derives the sizing equation for L_s in both converters, assuming a unity transformer turns-ratio ($n = 1$) throughout, and draws a direct comparison between the two approaches.

2 Dual Active Bridge (DAB)

2.1 Operating Principle

The DAB consists of two full-bridge voltage-source inverters connected through a high-frequency transformer and its series inductance L_s. The two bridges generate square-wave voltages at the

same switching frequency f_s, and power is transferred by introducing a controlled phase-shift ϕ (in radians) between the two bridge outputs. With Single-Phase-Shift (SPS) modulation and $n = 1$, the average transferred power is [1]:

$$P = \frac{V_1\, V_2}{2\pi f_s L_s} \cdot \phi\left(1 - \frac{|\phi|}{\pi}\right), \qquad |\phi| \le \frac{\pi}{2}. \tag{1}$$

2.2 Maximum Power and Inductance Sizing

Differentiating (1) with respect to ϕ shows that the power is maximised at $\phi = \pi/2$, giving:

$$P_{\max} = \frac{V_1\, V_2}{8\, f_s\, L_s}. \tag{2}$$

Inverting (2) yields the fundamental sizing equation:

$$\boxed{L_s = \frac{V_1\, V_2}{8\, f_s\, P_{\max}}.} \tag{3}$$

2.3 Design Considerations

Equation (3) is *closed-form*: given the DC bus voltages, the switching frequency, and the rated power, L_s is determined uniquely. Larger L_s reduces $P_{\max}$ and increases the phase-shift required at a given operating point, which can improve light-load ZVS margins but worsens reactive-current stress at full load. The relationship between power and phase-shift is *linear* in ϕ for small angles, and the converter behaves essentially as a controlled current source driven by a voltage difference.

3 CLLC Resonant Converter

3.1 Operating Principle

The CLLC is the bidirectional extension of the LLC resonant converter. The resonant network is formed by the series inductance L_r, the series capacitance C_r, and the magnetising inductance L_m on the primary side; a symmetric network (L_r', C_r') is placed on the secondary to preserve bidirectional resonant behaviour. The converter is operated close to the series resonant frequency:

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}. \tag{4}$$

3.2 First Harmonic Approximation (FHA)

Under FHA, the nonlinear switching network is replaced by sinusoidal sources and the load is represented by its equivalent AC resistance. For a resistive output load R_{out} and $n = 1$, the equivalent AC load resistance seen by the resonant tank is:

$$R_{ac} = \frac{8}{\pi^2} \cdot \frac{V_2^2}{P} = \frac{8}{\pi^2} \cdot R_{\text{out}}. \tag{5}$$

3.3 Quality Factor and Inductance Sizing

The characteristic impedance of the resonant tank is:

$$Z_r = \sqrt{\frac{L_r}{C_r}}, \tag{6}$$

and the loaded quality factor at resonance is:

$$Q = \frac{Z_r}{R_{ac}} = \frac{\sqrt{L_r/C_r}}{R_{ac}}. \tag{7}$$

From (4) and (7), solving for L_r and C_r:

$$\boxed{L_r = \frac{Q \cdot R_{ac}}{2\pi f_r},} \qquad C_r = \frac{1}{Q \cdot R_{ac} \cdot 2\pi f_r}. \tag{8}$$

A typical design targets $Q \leq 0.5 \div 1$ at full load in order to guarantee ZVS over an adequate load range.

3.4 Design Considerations

Unlike the DAB, the CLLC does not have a single closed-form equation that directly links L_r to $P_{\max}$. The role of L_r is to set the tank impedance Z_r and, together with L_m, to shape the converter voltage gain characteristic. The three key parameters—f_r, Q, and the inductance ratio $k = L_m/L_r$—must be chosen simultaneously to satisfy: (i) the required voltage-gain range; (ii) ZVS over the desired load range; and (iii) an acceptable peak-current stress. The primary control variable is the switching frequency f_s, which is varied around f_r to regulate the output voltage.

4 Direct Comparison

Table 1 summarises the principal differences between the two approaches.

Table 1: DAB vs. CLLC: comparison of the series inductance design.

Aspect	**DAB**	**CLLC**
Power-transfer mechanism	Phase-shift across L_s	Series resonance L_r–C_r
$P_{\max}$ sizing equation	$L_s = \dfrac{V_1 V_2}{8 f_s P_{\max}}$	$L_r = \dfrac{Q \cdot R_{ac}}{2\pi f_r}$
Primary control variable	Phase-shift ϕ at fixed f_s	Frequency ratio f_s/f_r
ZVS condition	Conditional (depends on ϕ, load)	Natural around f_r
Effect of larger L_s	Lower $P_{\max}$, higher ϕ at rated load	Higher Z_r, higher Q
Design complexity	Closed-form, single equation	Iterative, multi-parameter

5 Summary

For the DAB, the series inductance L_s is determined by a single, closed-form expression (3) that directly links the maximum transferable power to the DC bus voltages and the switching frequency. The design procedure is therefore straightforward.

For the CLLC, the resonant inductance L_r is obtained from (8) once the resonant frequency f_r and the target quality factor Q have been chosen. The choice of Q and of the inductance ratio $k = L_m/L_r$ is, however, inherently iterative because these parameters jointly determine the voltage-gain curve, the ZVS range, and the peak circulating current.

References

[1] R. W. De Doncker, D. M. Divan, and M. H. Kheraluwala, "A three-phase soft-switched high-power-density DC/DC converter for high-power applications," *IEEE Transactions on Industry Applications*, vol. 27, no. 1, pp. 63–73, Jan./Feb. 1991.

[2] R. Beiranvand, B. Rashidian, M. R. Zolghadri, and S. M. H. Alavi, "Using LLC resonant converter for designing wide-range voltage source," *IEEE Transactions on Industrial Electronics*, vol. 58, no. 5, pp. 1746–1756, May 2011.